SEVA FIRST INNOVATION CHALLENGE 2026

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TITLE PAGE

- **Problem Statement ID –** [PS ID]
- **Problem Statement Title-** RAKSHA-PATH X: Intelligent Emergency Route Optimization and Real-Time Disaster Road Intelligence System
- **Theme-** Surakshit Bharat (Theme 5)
- **PS Category-** Software
- **Team ID-** [Team ID]
- **Team Name (Registered on portal) -** RAKSHA-PATHX-100

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RAKSHA-PATH X

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Proof-of-Passability Emergency Intelligence Network · "Every route needs evidence. Every rescue needs a safer path."

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution

- A decision-support platform for emergency control rooms that rates every road segment by the evidence behind it: recency, source trust, verification
- Evidence-aware routing for ambulances, fire tenders and rescue units, with alternatives, vehicle restrictions and field-verification tasks

❖ How it addresses the problem

- Navigation apps assume roads are open. In floods and landslides they are not, and the cost is lost rescue time
- Never routes through a confirmed block; when no adequately verified route exists it says so and proposes the nearest reachable meeting point

❖ Innovation and uniqueness of the solution

- Proof-of-Passability: evidence expires per hazard type (flood 2 h, fallen tree 12 h); "no report" is treated as Unknown, never "open"
- Transparent "Why this status?" trail instead of a black-box AI score; offline-first; humans make the final call

Every road gets an evidence-based status

- **Verified passable**
  Recent evidence from an authenticated or verified source
- **Potential hazard**
  Hazard reported, not yet verified
- **Blocked**
  Confirmed; never used for routing
- **Unknown**
  No adequate evidence (no report is not 'open')
- **Stale evidence**
  Evidence older than its hazard-specific limit
- **Conflicting evidence**
  Reports disagree; both kept, verifier tasked

TECHNICAL APPROACH

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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

Frontend: React 18 + TypeScript, Vite, Tailwind CSS (responsive, keyboard-accessible)

Map: Custom SVG road-network renderer; no paid map or API keys

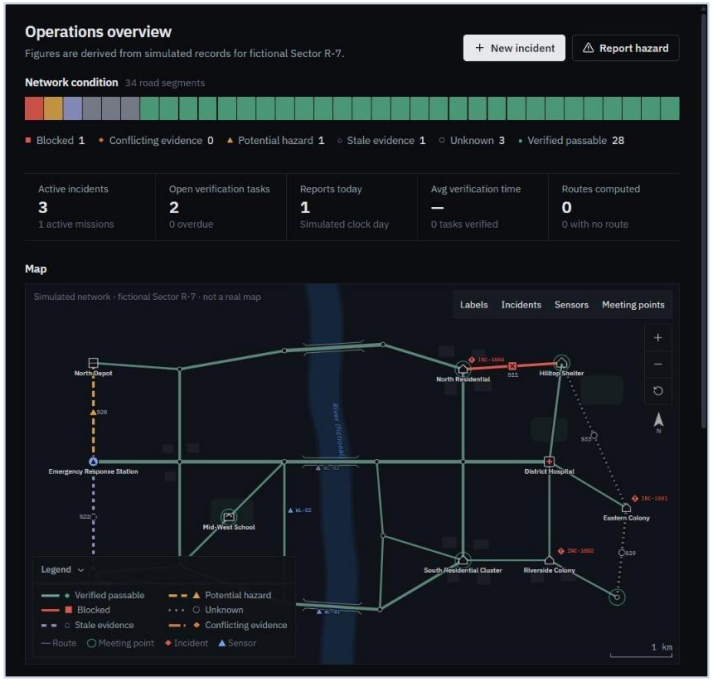
Routing: Dijkstra graph engine in TypeScript: 3 modes, alternatives, last-mile

Evidence: Rule engine: freshness, trust tiers, duplicates, conflicts

Data: LocalStorage persistence, Service Worker offline cache, CSV export

Quality: Vitest: 34 automated tests, 27 evaluation cases, 8 demo scenarios

Hardware (opt.): ESP32 + water-level sensor sending {sensorId, valueCm, ts} over Wi-Fi



Working prototype: operations dashboard (simulated data)

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)

Inputs
Crowd reports, field inspections, sensor readings

→

Evidence engine
Freshness · trust tier · duplicates · conflicts

→

Road status
6 statuses + “Why this status?” explanation

→

Routing
Fastest · Evidence-prioritized · Conservative

→

Decision output
Route + alternatives, or “no verified route” + meeting point

→

Field verification
Verifier confirms; result updates graph and reroutes

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FEASIBILITY AND VIABILITY

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❖ Analysis of the feasibility of the idea

Working	prototype built and demonstrable today
₹0	software licences, API keys or paid maps
~₹2,600	per optional sensor node (estimate*)
27/27	evaluation cases pass in the test suite

Runs offline on an ordinary laptop; open-source stack; one-click Windows launcher.

**Indicative component estimates; require supplier verification.*

Potential challenges and risks	Strategies for overcoming these challenges
False or prank crowd reports	Trust tiers; anonymous reports stay unverified; duplicate detection; verification tasks
Outdated information	Hazard-specific expiry; stale roads flagged and queued for re-inspection
Network loss during disasters	Offline-first: reports queued, synced later with conflict detection
Over-reliance on automation	Decision support only; human confirmation; never labels a route “safe”
Sensor faults or misreadings	A sensor alone never marks a road passable; out-of-range and disconnect alerts
Real road data & approvals	Pilot with local authority GIS / OpenStreetMap data under formal permission

IMPACT AND BENEFITS

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❖ Potential impact on the target audience

- District Emergency Operation Centres and control rooms
- Ambulance, fire and rescue crews (SDRF / NDRF teams)
- Local authorities and ward offices verifying roads
- Residents of flood- and landslide-prone areas

Measured in the prototype vs. a plain shortest-path baseline

4 → 0	7 → 0	8 / 8
confirmed-blocked road segments routed through	unverified segments on chosen routes	end-to-end demo scenarios pass

11 routing test cases on simulated data; not field validation.

❖ Benefits of the solution (social, economic, environmental, etc.)

Social
Fewer rescue vehicles sent into blocked or flooded roads; faster verified access to people in need

Economic
Zero licence cost; runs on existing laptops and phones; low-cost optional sensors

Environmental
Fewer wasted detours and idle trips; reusable flood-level data for planning

Governance
Every decision explainable and logged; supports accountable, auditable response

RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

Research basis for the design

- Consumer navigation optimises travel time from traffic data; it does not certify that a road is physically passable during floods or landslides
- Disaster reports are often crowd-sourced, unverified and quickly outdated, so evidence needs freshness and trust levels
- Emergency decisions must be explainable and auditable, so we use transparent rules, not black-box scores
- Connectivity degrades in disasters, so the system has to work offline first

References

1. NDMA, National Disaster Management Plan (2019) https://ndma.gov.in

2. NDMA, Guidelines on Management of Urban Flooding (2010) https://ndma.gov.in

3. UNDRR, Sendai Framework for Disaster Risk Reduction 2015–2030 https://www.undrr.org

4. Central Water Commission, flood forecasting services https://cwc.gov.in

5. Dijkstra, E. W. (1959). A note on two problems in connexion with graphs. Numerische Mathematik 1, 269–271

6. Humanitarian OpenStreetMap Team, open map data for disaster response https://www.hotosm.org

7. [Add any other papers / articles the team consulted]

Prototype

Source code: https://github.com/Abishekh2007/sewa-

Demo video: [Demo video URL]